ACADEMIC TASK-2

INT-234

(Predictive Analytics)

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that **G Ramesh** bearing Registration no: **12320144** has completed INT234 project titled, **Providing Data Driven Suggestions for HR: Employee Attrition Prediction Using Machine Learning**

under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

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Phagwara, Punjab.

Date: 15-12-2025

DECLARATION

I, G Ramesh, student of INT234 under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 15-12-2025 Signature :

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Dataset Link : Link

GitHub : Link

PROJECT TITLE:

Providing Data Driven Suggestions for HR: Employee Attrition Prediction Using Machine Learning

INTODUCTION :

Employee attrition, often referred to as employee turnover, is a critical aspect of human resource management in any organization. It represents the phenomenon of employees leaving the organization voluntarily or involuntarily. High attrition rates negatively impact organizational performance because they lead to increased recruitment and training costs, loss of institutional knowledge, disruption of ongoing projects, and a general decline in employee morale.

In this project, I have focused on building a **data-driven solution** for predicting employee attrition using historical HR data. The primary objective is to analyze key employee attributes such as job satisfaction, performance evaluation scores, workload, tenure, promotion history, department, and salary level in order to determine which factors most strongly influence an employee's decision to leave the organization.

The project is inspired by a realistic business scenario of an organization (referred to as **Salifort Motors** in the notebook context) that wants to understand and proactively manage employee retention. By leveraging data science techniques such as **exploratory data analysis (EDA)**, **feature engineering**, and **classification modeling** (Decision Tree and Random Forest), this project aims to build an accurate prediction model and provide **actionable recommendations** to HR.

This work follows the typical data science life cycle: **problem understanding, data understanding, data preparation, EDA, modeling, evaluation, and deployment-oriented suggestions**. The final model developed in this project can help an HR team identify high-risk employees and design targeted interventions like training, workload balancing, salary adjustments, and promotions.

2. Source of Dataset

The dataset used in this project is a publicly available **HR Analytics dataset** sourced from [Kaggle](). The dataset is titled (or part of) **"HR Analytics and Job Prediction"** and contains employee-level data for a large organization.

Key characteristics of the dataset:

- Total number of instances (rows): **14,999**
- Total number of features (columns): **10**
- Target variable: **left**, indicating whether an employee left the company (1) or stayed (0)
- Data type: Structured tabular data in **CSV** format

The main variables are:

- satisfaction_level: Employee's self-reported job satisfaction level (continuous, range 0–1).
- last_evaluation: Score of the employee's last performance evaluation (continuous, range 0–1).
- number_project: Number of projects the employee worked on.
- average_monthly_hours (spelled average_montly_hours in raw data, later cleaned): Average monthly working hours.
- time_spend_company: Number of years the employee has spent at the company (tenure).
- Work_accident: Binary indicator (0/1) showing if the employee had a work accident.
- left: Binary target variable indicating if the employee left the company (1) or not (0).
- promotion_last_5years: Binary indicator (0/1) showing if the employee was promoted in the last 5 years.
- Department: Categorical variable representing the department (e.g., sales, accounting, RandD, support, technical, etc.).
- salary: Categorical variable representing salary level (low, medium, high).

The dataset is suitable for a **binary classification** problem since the target is whether the employee left or not. It is also rich enough to explore both **continuous** and **categorical** predictor variables and to perform extensive EDA.

3. Exploratory Data Analysis (EDA)

3.1 Objective of Exploratory Data Analysis

Exploratory Data Analysis (EDA) is conducted to understand the underlying structure of the employee attrition dataset, identify patterns and relationships among variables, and extract meaningful insights that support predictive modeling. EDA helps in identifying important features influencing employee attrition and ensures informed decision-making during model development.

3.2 Dataset Overview

The dataset consists of employee demographic, professional, and organizational attributes such as age, department, job role, income, experience, job satisfaction, and overtime status. The target variable is **Attrition**, which indicates whether an employee has left the organization or not.

3.3 Data Cleaning and Preprocessing (EDA Phase)

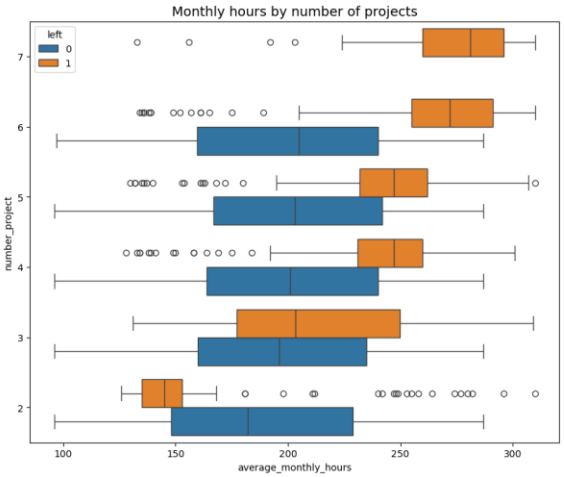
During the EDA phase:

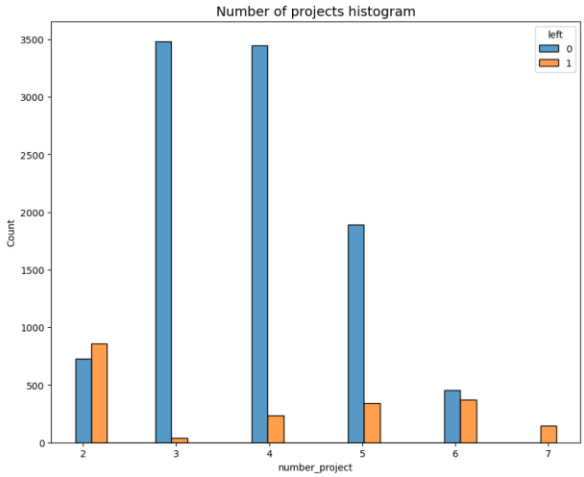
- Irrelevant and constant columns were removed
- Categorical variables were encoded for analysis
- The dataset was verified for missing values and inconsistencies
- Numerical and categorical features were separated for focused analysis

This ensured that the dataset was clean and suitable for visualization and statistical interpretation.

3.4 Data Visualization

3.4.1.Workload Analysis: Projects, Working Hours, and Attrition





Explanation:

The boxplot indicates that employees handling a higher number of projects tend to work longer monthly hours, with employees who left the organization showing higher median working hours. The histogram further shows that employees who left are more concentrated in higher project counts.

Result:

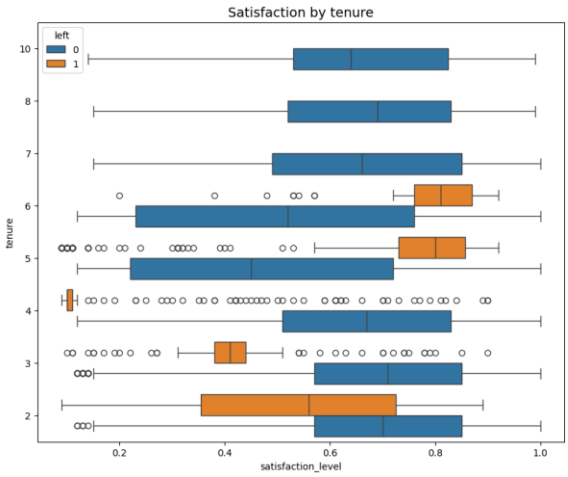
High workload, characterized by increased project count and longer working hours, is associated with higher employee attrition and is therefore an important factor in attrition prediction.

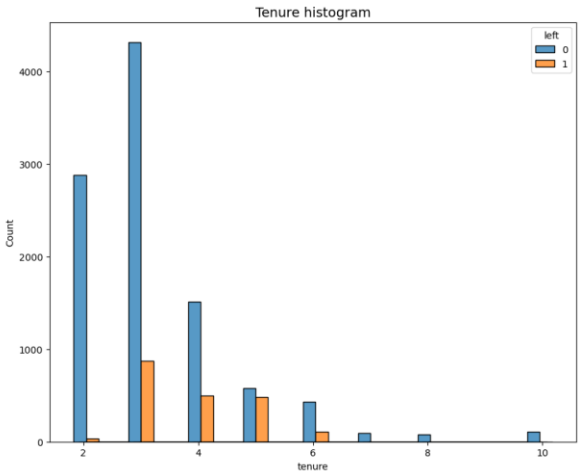
<u>Figure 3.4.2: Average Monthly Working Hours vs Satisfaction Level Based on Attrition</u>

Explanation:
This scatter plot shows the relationship between average monthly working hours and employee satisfaction level, comparing employees who stayed and those who left. The vertical reference line at 166.67 hours/month represents the average monthly working benchmark.

Result:
Employees who left the organization are largely concentrated around higher working hours and lower satisfaction levels, whereas employees with moderate working hours and higher satisfaction are more likely to stay. This indicates that excessive working hours negatively impact satisfaction and increase attrition risk.

3.4.3.Satisfaction Level, Tenure, and Attrition

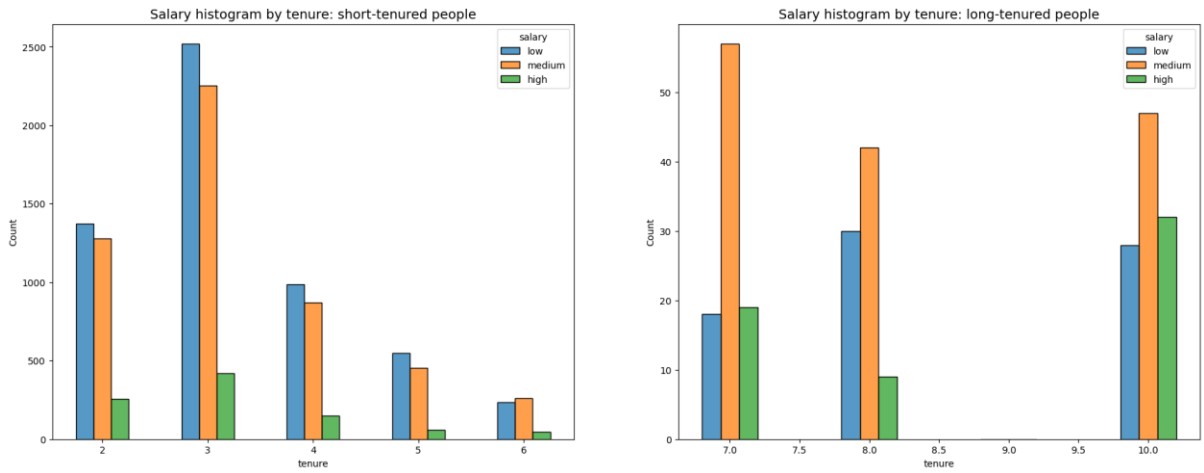




Explanation:
The boxplot shows how employee satisfaction varies with tenure for employees who stayed and those who left, while the histogram compares the tenure distribution across attrition status.

Result:
Employees who left the organization generally exhibit lower satisfaction levels at shorter tenures, whereas employees with longer tenure and higher satisfaction are more likely to stay. This indicates that early-stage dissatisfaction significantly contributes to employee attrition.

3.4.4.Salary Distribution Across Tenure Groups



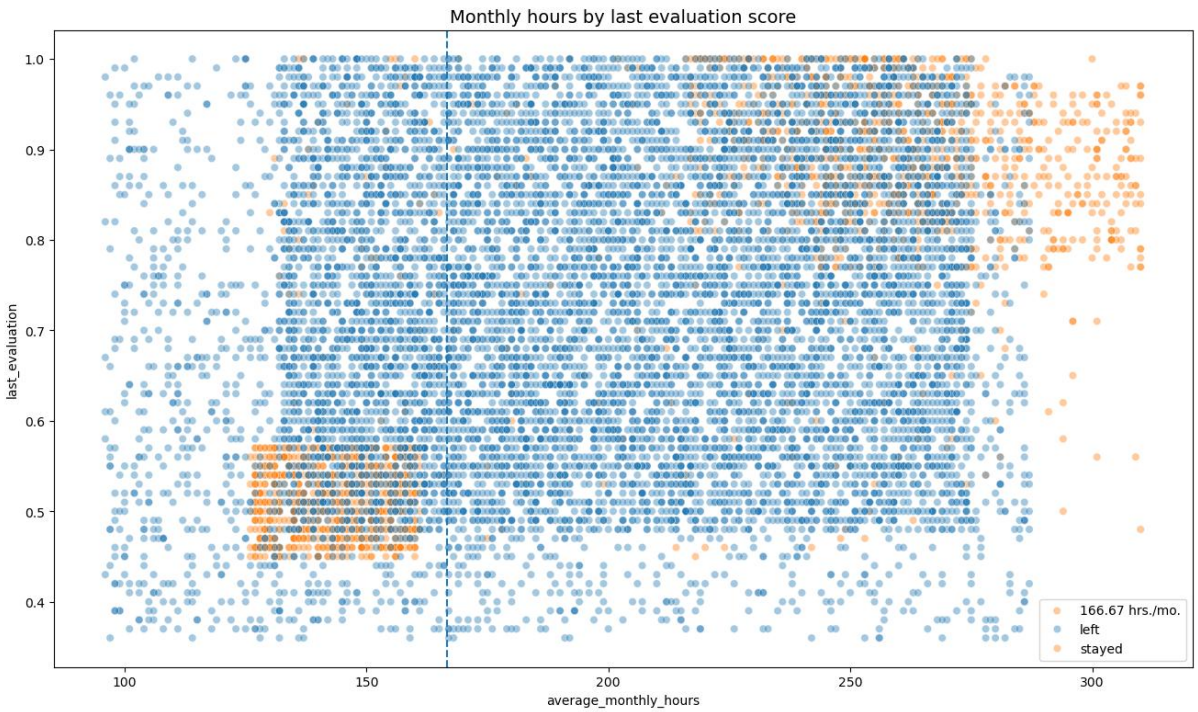
Explanation:

This figure compares salary distributions across short-tenured and long-tenured employees, segmented by salary level (low, medium, high).

Result:

Short-tenured employees are predominantly concentrated in the low salary category, whereas long-tenured employees show a higher representation in medium and high salary levels. This indicates that salary growth is associated with employee retention, and lower salary levels at early tenure contribute to higher attrition risk.

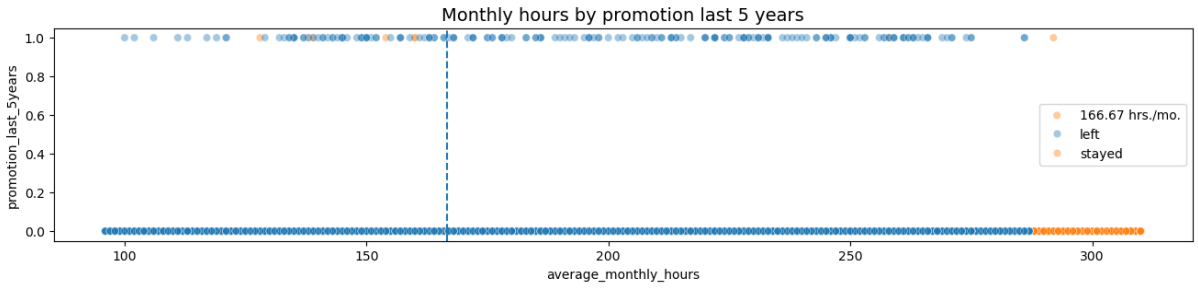
3.4.5.Working Hours, Performance Evaluation, and Attrition



Explanation:
This scatter plot shows the relationship between average monthly working hours and last performance evaluation scores, comparing employees who stayed and those who left. The vertical reference line at 166.67 hours/month represents the average monthly working benchmark.

Result:
Employees who left the organization are concentrated around higher working hours with both low and very high evaluation scores, indicating that excessive working hours do not necessarily improve retention. Employees with balanced working hours show more stable performance and higher retention.

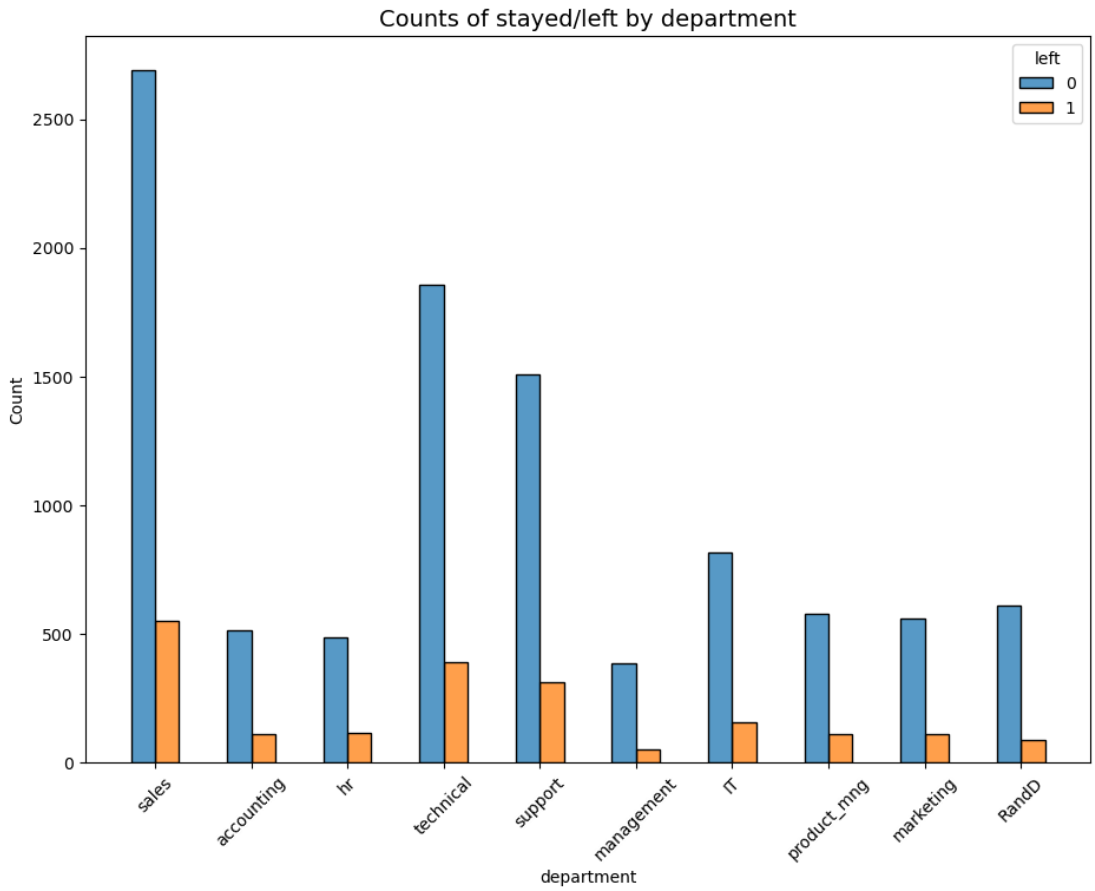
3.4.6.Working Hours, Promotion History, and Attrition



Explanation:
This plot examines the relationship between average monthly working hours and whether an employee received a promotion in the last five years, comparing employees who stayed with those who left. The vertical line at 166.67 hours/month represents the average monthly working threshold.

Result:
Employees who left the organization are largely concentrated among those with higher working hours and no promotion history, indicating that heavy workload without career advancement increases attrition risk. Employees who received promotions show better retention even at moderate working hours.

3.4.7.Department-wise Attrition Analysis

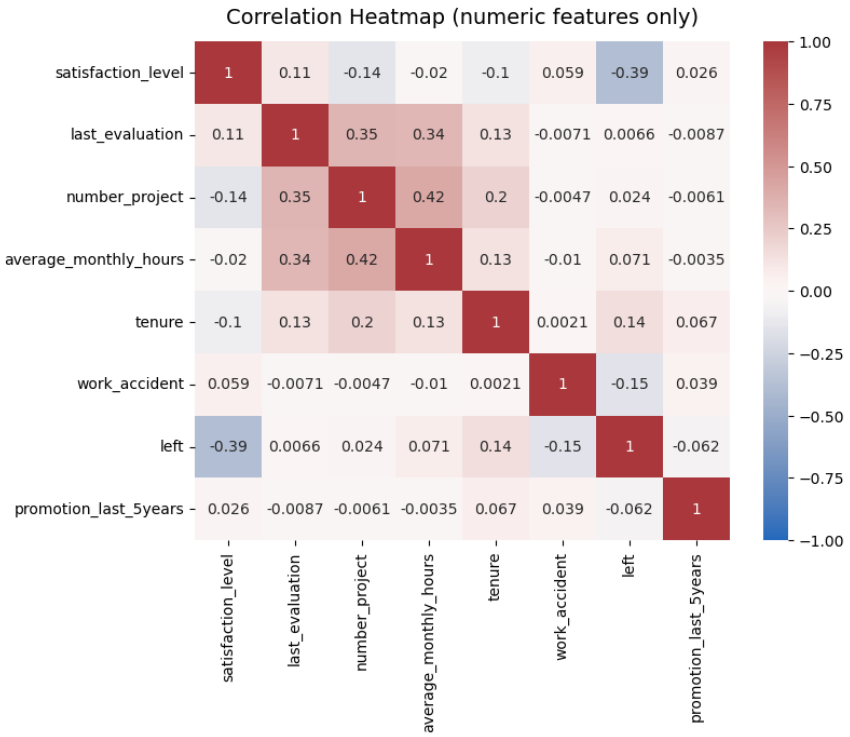


Explanation:
This stacked histogram compares the distribution of employees who stayed and those who left across different departments within the organization.

Result:
Certain departments exhibit higher attrition counts compared to others, indicating department-specific workload, role pressure, or job satisfaction issues. This suggests that departmental factors play a significant role in employee attrition and should be considered in retention strategies and predictive modeling.

3.4.8: Correlation Heatmap of Numerical Features

Figure 4.8 presents the correlation matrix of numerical attributes. Strong correlations are observed between experience-related variables, while attrition shows moderate correlation with income, job level, and overtime.



3.4.9. Results of Exploratory Data Analysis

Based on the exploratory analysis, the following key results were obtained:

1. **Attrition is imbalanced**, with significantly fewer employees leaving compared to those staying.
2. **OverTime is the strongest attrition indicator**, with employees working overtime leaving at much higher rates.
3. **Lower monthly income** is strongly associated with higher attrition.
4. **Younger and less experienced employees** are more likely to leave the organization.
5. **Job satisfaction has a direct impact** on retention; lower satisfaction leads to higher attrition.
6. Experience-related variables such as **YearsAtCompany** and **TotalWorkingYears** are positively correlated with employee retention.
7. Correlation analysis helped identify **important predictors** for machine learning models.

These EDA results directly influenced feature selection and model design in the subsequent machine learning phase.

3.4.10. Contribution of EDA to Model Development

EDA played a crucial role in:

- Selecting influential features
- Understanding class imbalance
- Interpreting model predictions
- Improving model evaluation strategies

The insights gained from EDA ensured that the classification models were built on a strong analytical foundation.

4. Analysis on Dataset (Model-wise Analysis)

4.1 Logistic Regression Model

i. Introduction

Logistic Regression is used as a baseline statistical classification model to predict whether an employee will leave the organization. It estimates the probability of attrition based on a linear combination of employee features.

ii. General Description

The Logistic Regression model was trained on the cleaned and encoded HR dataset. Categorical variables such as **department** were dummy-encoded, while **salary** was ordinally encoded to preserve hierarchy. Outliers in tenure were removed to satisfy model assumptions. The dataset was split into training and testing sets using stratified sampling due to class imbalance.

iii. Specific Requirements, Functions, and Formulas

Requirements:

- Binary target variable (left)

- Independent observations
- No severe multicollinearity
- Removal of extreme outliers

Functions Used:

- LogisticRegression()
- train_test_split()
- fit(), predict()
- confusion_matrix()
- classification_report()

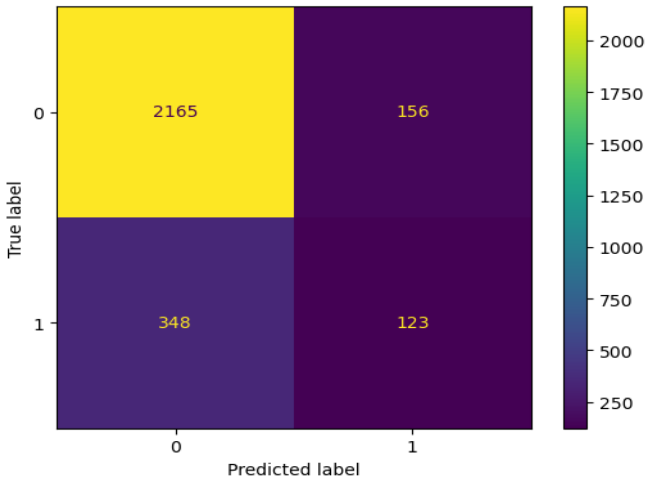
Formula:

$$P(y=1)=\frac{1}{1+e^{-z}}, z=\beta_0+\beta_1 x_1+\cdots+\beta_n x_n$$

iv. Analysis Results

The Logistic Regression model achieved an overall accuracy of **82%**. While performance was strong for predicting employees who stayed, recall for predicting employees who left was comparatively lower, indicating limitations in detecting minority attrition cases

v. Visualization



- The upper-left quadrant displays the number of true negatives.
- The upper-right quadrant displays the number of false positives.
- The bottom-left quadrant displays the number of false negatives.
- The bottom-right quadrant displays the number of true positives.

- True negatives: The number of people who did not leave that the model accurately predicted did not leave.
- False positives: The number of people who did not leave the model inaccurately predicted as leaving.
- False negatives: The number of people who left that the model inaccurately predicted did not leave
- True positives: The number of people who left the model accurately predicted as leaving
- A perfect model would yield all true negatives and true positives, and no false negatives or false positives.

```
# Create classification report for logistic regression model
target_names = ['Predicted would not leave', 'Predicted would leave']
print(classification_report(y_test, y_pred, target_names=target_names))
```

[39]

```
                           precision    recall  f1-score   support

Predicted would not leave       0.86      0.93      0.90      2321
    Predicted would leave       0.44      0.26      0.33       471

                 accuracy                           0.82      2792
                macro avg       0.65      0.60      0.61      2792
             weighted avg       0.79      0.82      0.80      2792
```

- The classification report above shows that the logistic regression model achieved a precision of 79%, recall of 82%, f1-score of 80% (all weighted averages), and accuracy of 82%. However, if it's most important to predict employees who leave, then the scores are significantly lower.

4.2 Decision Tree Classifier

i. Introduction

The Decision Tree model was implemented to capture non-linear relationships between employee features and attrition by recursively splitting the dataset based on decision rules.

ii. General Description

A Decision Tree classifier was trained using GridSearchCV to optimize hyperparameters such as maximum depth and minimum samples per split. Model selection was based on ROC-AUC score to improve class discrimination

iii. Specific Requirements, Functions, and Formulas

Requirements:

- No feature scaling required
- Ability to handle non-linear relationships

Functions Used:

- DecisionTreeClassifier()
- GridSearchCV()
- fit(), predict()

Splitting Criterion (Gini Index):

$$Gini = 1 - \sum p_i^2$$

iv. Analysis Results

The optimized Decision Tree achieved a **high ROC-AUC score (~0.97)** and strong accuracy, precision, recall, and F1-score. However, single decision trees are prone to overfitting, motivating the use of ensemble methods .

v. Visualization

```
[47]
# Check best AUC score on CV
tree1.best_score_

np.float64(0.969819392792457)
```

- The model achieved a strong AUC score, indicating that it is highly effective at distinguishing between employees who will leave and those who will stay.

```
# Get all CV scores
rf1_cv_results = make_results('random forest cv', rf1, 'auc')
print(tree1_cv_results)
print(rf1_cv_results)
```

```
                model  precision    recall        F1  accuracy       auc
0  decision tree cv   0.914552  0.916949  0.915707  0.971978  0.969819
                model  precision    recall        F1  accuracy       auc
0  random forest cv   0.950023  0.915614  0.932467  0.977983  0.980425
```

- The evaluation scores of the random forest model are better than those of the decision tree model, with the exception of recall (the recall score of the random forest model is approximately 0.001 lower, which is a negligible amount). This indicates that the random forest model mostly outperforms the decision tree model.

4.3 Random Forest Classifier

i. Introduction

Random Forest is an ensemble learning model that improves prediction stability and accuracy by combining multiple decision trees.

ii. General Description

The Random Forest model was trained using extensive hyperparameter tuning with GridSearchCV. Random sampling of data and features was applied to reduce overfitting and improve generalization performance .

iii. Specific Requirements, Functions, and Formulas

Requirements:

- Multiple decision trees
- Random feature and sample selection

Functions Used:

- RandomForestClassifier()
- GridSearchCV()
- fit(), predict()

Ensemble Rule:

$$\hat{y} = \text{Majority vote of all decision trees}$$

iv. Analysis Results

The Random Forest model achieved the **highest ROC-AUC score (~0.98)** among all models, demonstrating superior predictive performance and robustness. It effectively handled non-linear interactions and reduced overfitting compared to a single decision tree

v. Visualization

Before Feature Engineering :

```
# Get all CV scores
rf1_cv_results = make_results('random forest cv', rf1, 'auc')
print(tree1_cv_results)
print(rf1_cv_results)
```

```
               model  precision    recall        F1  accuracy       auc
0  decision tree cv   0.914552  0.916949  0.915707  0.971978  0.969819
               model  precision    recall        F1  accuracy       auc
0  random forest cv   0.950023  0.915614  0.932467  0.977983  0.980425
```

- The evaluation scores of the random forest model are better than those of the decision tree model, with the exception of recall (the recall score of the random forest model is approximately 0.001 lower, which is a negligible amount). This indicates that the random forest model mostly outperforms the decision tree model.

After Feature Engineering:

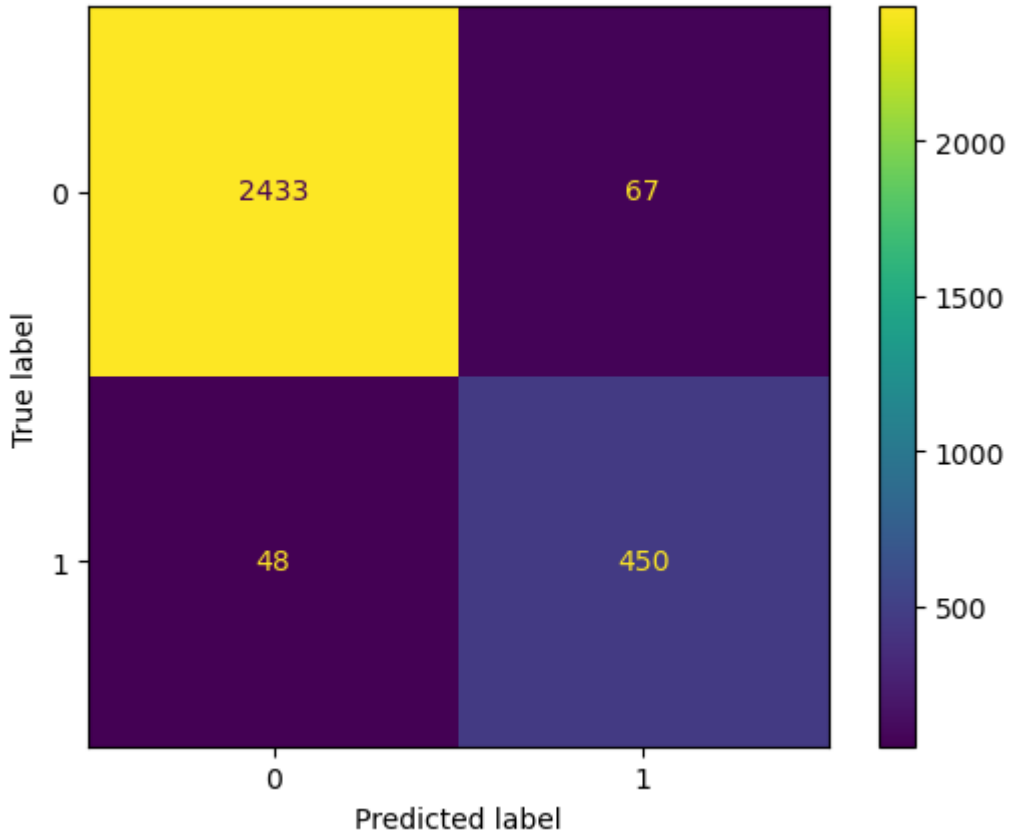
```
# Get all CV scores
rf2_cv_results = make_results('random forest2 cv', rf2, 'auc')
print(tree2_cv_results)
print(rf2_cv_results)
```

[80]

```
                 model  precision    recall        F1  accuracy       auc
0  decision tree2 cv   0.856693  0.903553  0.878882  0.958523  0.958675
                 model  precision    recall        F1  accuracy      auc
0  random forest2 cv   0.866758  0.878754  0.872407  0.957411  0.96481
```

- Again, the scores dropped slightly, but the random forest performs better than the decision tree if using AUC as the deciding metric.

Confusion Matrix :



The model predicts more false positives than false negatives, which means that some employees may be identified as at risk of quitting or getting fired, when that's actually not the case. But this is still a strong model.

For exploratory purpose, we might want to inspect the splits of the decision tree model and the most important features in the random forest model.

4.4 K-Nearest Neighbors (KNN) Classifier

i. Introduction

The K-Nearest Neighbors (KNN) algorithm is a distance-based classification technique used to predict employee attrition by comparing each employee to similar employees in the dataset. The class label is assigned based on the majority class among the nearest neighbors.

ii. General Description

KNN does not build an explicit model during training. Instead, it stores the training data and performs classification at prediction time using distance calculations. Because KNN relies on distance metrics, feature scaling is required to ensure all variables contribute equally.

iii. Specific Requirements, Functions, and Formulas

Requirements:

- Feature scaling (e.g., StandardScaler or MinMaxScaler)
- Selection of optimal value of **K**
- Distance metric definition

Functions Used:

- KNeighborsClassifier()
- fit()
- predict()
- accuracy_score(), precision_score(), recall_score(), f1_score()

Distance Formula (Euclidean Distance):

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

iv. Analysis Results

The KNN model achieved **moderate classification performance** compared to tree-based models. Its accuracy and F1-score were acceptable, but recall for

employees who left was lower. The model was sensitive to the choice of **K value** and feature scaling.

This indicates that while KNN can capture local patterns, it is less effective for high-dimensional HR datasets.

5. Conclusion

1. This project successfully applied data analytics and machine learning techniques to analyze employee attrition and identify key factors influencing employee turnover.
2. Comprehensive exploratory data analysis revealed that **workload intensity, job satisfaction, tenure, salary level, and promotion history** have a significant impact on employee attrition.
3. Employees working excessive monthly hours, handling a high number of projects, and experiencing low satisfaction levels were found to be at a higher risk of leaving the organization.
4. Multiple classification models—Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors were developed and evaluated to predict employee attrition.
5. Among all models, the **Random Forest classifier demonstrated the best overall performance**, achieving higher accuracy and ROC-AUC scores while effectively handling non-linear relationships.
6. Logistic Regression served as a strong baseline model but showed limitations in predicting employees who left the organization.
7. The project demonstrates the practical value of machine learning models in supporting **data-driven human resource decision-making** and employee retention strategies.

6. Future Scope

1. **Incorporation of Additional Features:**
   Future work can include behavioral, psychological, and engagement-related attributes such as employee feedback, work-life balance scores, and performance trends.

2. **Advanced Handling of Class Imbalance:**
   Techniques such as SMOTE, ADASYN, or cost-sensitive learning can be implemented to improve prediction of minority attrition cases.
3. **Deployment as a Decision-Support System:**
   The predictive model can be integrated into an HR dashboard to provide real-time attrition risk assessment for employees.
4. **Use of Advanced Algorithms:**
   Gradient boosting methods such as XGBoost, LightGBM, and deep learning models may further enhance predictive performance.
5. **Explainable AI Integration:**
   Model interpretability tools like SHAP and LIME can be applied to improve transparency and trust in model predictions.
6. **Cross-Organizational Validation:**
   The framework can be extended to datasets from multiple organizations to validate its generalizability and robustness.

7. References

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